eg. No. | | | | | | | | | | |

[12ME62/13ME62]

SIXTH SEMESTER B.Tech./B.E. EXAMINATION, JUNE – 2016

HEAT AND MASS TRANSFER

Time: 3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full Questions.
2. Missing data, if any, may be suitably assumed.
3. Use of Heat Transfer Data Hand Book permitted.

1. a. Derive the general heat conduction equation in Cartesian coordinates for an isotropic material. Mention the assumptions made. **06**

b. A steam pipe of 5cm inner diameter and 6.5 cm outer diameter is insulated with a 2.75 cm radial thickness of high temperature insulation having k=1.1 W/mK. Inside and outside heat transfer coefficients are 4650 W/m^{2}K and 11.5 W/m^{2}K. Thermal conductivity of pipe material is 45 W/mK: If the steam temperature is 200°C and ambient temperature is 20°C, determine. **08**

i. Heat loss per meter length of pipe.

ii. Temperatures at the interface.

iii. Overall heat transfer coefficient based on outer surface area.

2. a. Define critical thickness of insulation. Obtain an expression for the critical radius of insulation of cylinder. **08**

b. Twelve thin brass fins (k=76 W/mK) of 0.8 mm thick are placed longitudinally on a 5cm diameter cylinder which stands vertically and is surrounded by air at 40°C. The fins extend 5.5 cm from the cylinder surface. Heat transfer coefficient on the fin surface is 24 W/m^{2}K. Evaluate the total rate of heat transfer to the air per meter length of the cylinder when its surface is at 150°C. Take $\theta_0 = 110°C$. **06**

3. a. Derive an expression for the instantaneous and total heat flow in terms of the product of Biot and Fourier number in one dimensional transient heat conduction. **06**

b. A 15mm diameter mild steel sphere (k=42 W/mK) is exposed to cooling air flow at 20°C resulting in convective coefficient 120 W/m^{2}K. Evaluate the following: **08**

i. Time required to cool the sphere from 550°C to 90°C.

ii. Instantaneous heat transfer rate 2 minutes after the start of cooling

iii. Total energy transferred from the sphere during the first two minutes.

For mild steel, assume

$\rho = 7850 kg/m^3;\ c_p = 475 J/kg\ k$

$\alpha = 0.045\ m^2/hr$

4. a. Using dimensional analysis, obtain a relation between Nu, Re and Pr for Forced convection heat transfer. **06**

b. Air at 20^0C flows over a flat plate at 3m/s. Plate is heated uniformly throughout its entire length and is maintained at a surface temperature of 40°C. Calculate the distance from the leading edge at which the flow changes from laminar to turbulent conditions. Assume that the transition occurs at a critical Reynolds number of 5×10^5. Evaluate **08**

i. Thickness of hydrodynamic and thermal boundary layers at transition point

ii. Local and average heat transfer coefficient

iii. convective heat flow from plate to ambient air

consider unit width of the plate

5. a. Water flows through a 5 cm diameter tube at a velocity of 1m/s. Determine the heat lost in a length of 2m if the inlet temperature of water is 40°C and the outlet temperature of water is 80°C. The tube surface temperature is 100°C. **06**

b. A horizontal pipe 0.3m in diameter is maintained at a temperature of 250°C in a room where the ambient air is at 15°C. Calculate the free convection heat loss per meter length. **08**

6. a. With a neat sketch, explain the various regimes in boiling heat transfer. **06**

b. Write a note on film wise and drop wise condensation. **04**

c. State and explain Fick's law of diffusion. **04**

7. a. Define **03**

i. Heat exchanger

ii. Fouling factor

iii. Heat exchanger effectiveness

b. Derive an expression for LMTD of a parallel flow heat exchanger. State the assumptions made. **05**

c. In a counter flow double pipe heat exchanger, water is heated from 25°C to 65°c by oil with a specific heat of 1.45 KJ/kg K and mass flow rate of 0.9 kg/s. oil is cooled from 230^0C to 160°C. If overall heat transfer coefficient is 420 W/m^2 K, evaluate **06**

i. Rate of heat transfer

ii. Mass flow rate of water

iii. surface area of heat exchanger

8. **a.** State **10**

- **i.** Stefan-Boltzmann's Law
- **ii.** Kirchoff's law
- **iii.** Planck's law
- **iv.** Wein's displacement law
- **v.** Lambert's law

b. With reference to thermal radiation explain **04**

- **i.** absorptivity
- **ii.** reflectivity
- **iii.** transmissivity
- **iv.** radiosity